SIMATS ENGINEERING
CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems

COURSE CODE: ECA5604

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COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)
2. Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)
3. Simulation of beamforming gain under steering angle, antenna elements, frequency, SNR, and interference (BL4)
4. Simulation of subcarrier spacing effect under 15 kHz, 30 kHz, 60 kHz, 120 kHz, and symbol duration (BL4)
5. Simulation of propagation loss under 700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz (BL4)

Code of Conduct:

*I **<u>K.Poorna chandu(192312585)</u>**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: K.Poorna chandu.

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

1. Simulation of LTE Throughput Performance under varying SNR.

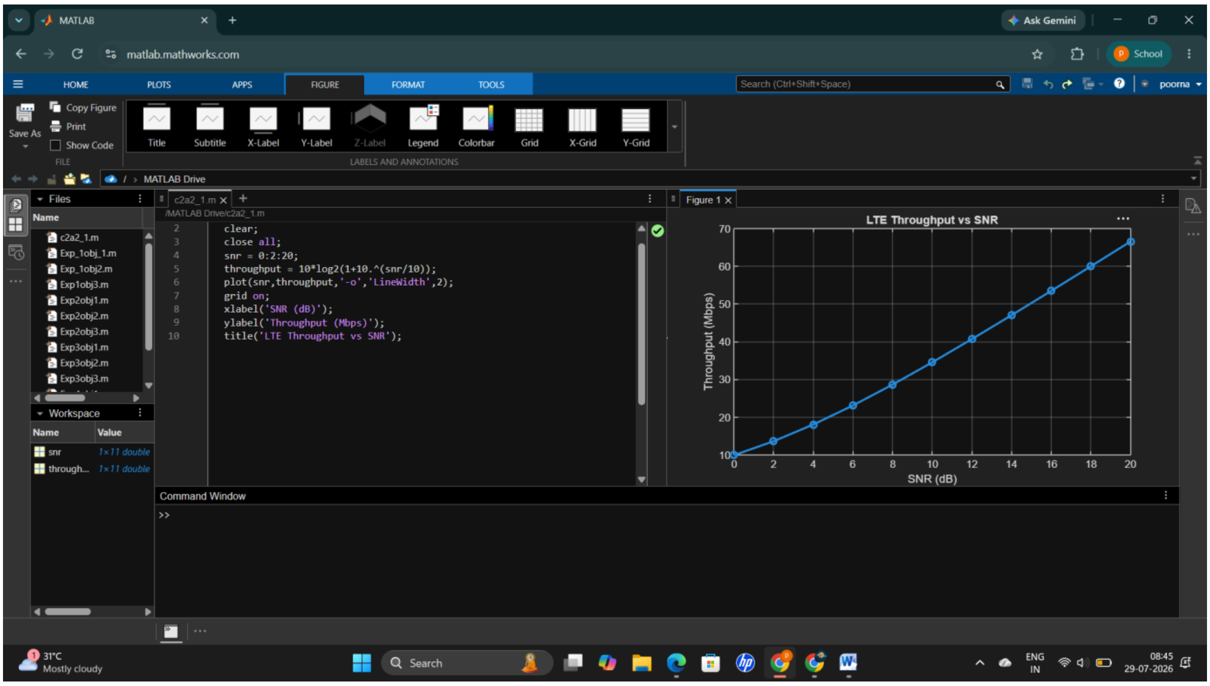
Aim:-

To simulate the LTE throughput performance for different Signal-to-Noise Ratio (SNR) values using MATLAB.

MATLAB Code:-

```
clear;
close all;
snr = 0:2:20;
throughput = 10*log2(1+10.^(snr/10));
plot(snr,throughput,'-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Throughput (Mbps)');
title('LTE Throughput vs SNR');
```

output:-



Result:-

The LTE throughput increases with increasing SNR, indicating better channel quality and higher data transmission rates.

2. **Simulation of BER Performance in MIMO under varying SNR.**

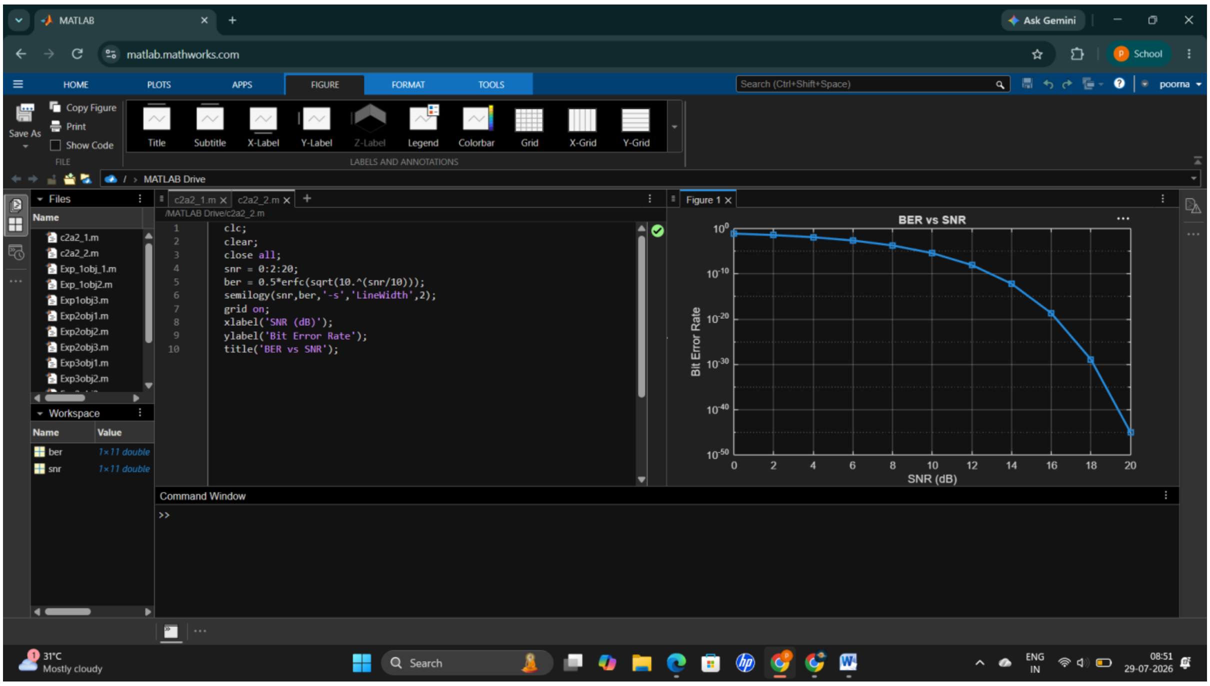
Aim:-

To simulate the Bit Error Rate (BER) performance under different SNR values.

MATLAB Code:-

```
clc;
clear;
close all;
snr = 0:2:20;
ber = 0.5*erfc(sqrt(10.^(snr/10)));
semilogy(snr,ber,'-s','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate');
title('BER vs SNR');
```

output:-



Result:-

BER decreases with increasing SNR, resulting in more reliable communication.

3. Simulation of Beamforming Gain.

Aim:-

To simulate beamforming gain for different steering angles.

MATLAB Code:-

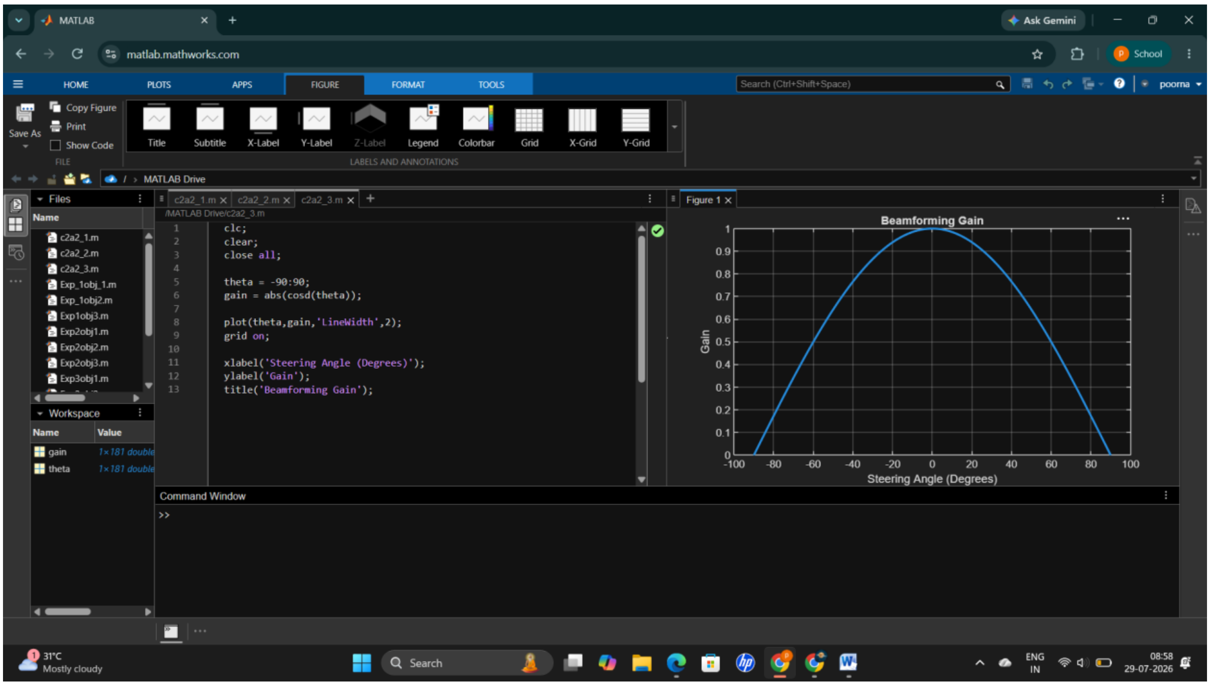
```
clc;
clear;
close all;

theta = -90:90;
gain = abs(cosd(theta));

plot(theta,gain,'LineWidth',2);
grid on;

xlabel('Steering Angle (Degrees)');
ylabel('Gain');
title('Beamforming Gain');
```

output:-



Result:-

The beamforming gain is maximum in the desired direction and decreases as the steering angle moves away from the center.

4. Simulation of Subcarrier Spacing Effect.

Aim:-

To compare different subcarrier spacing values used in 5G NR.

MATLAB Code:-

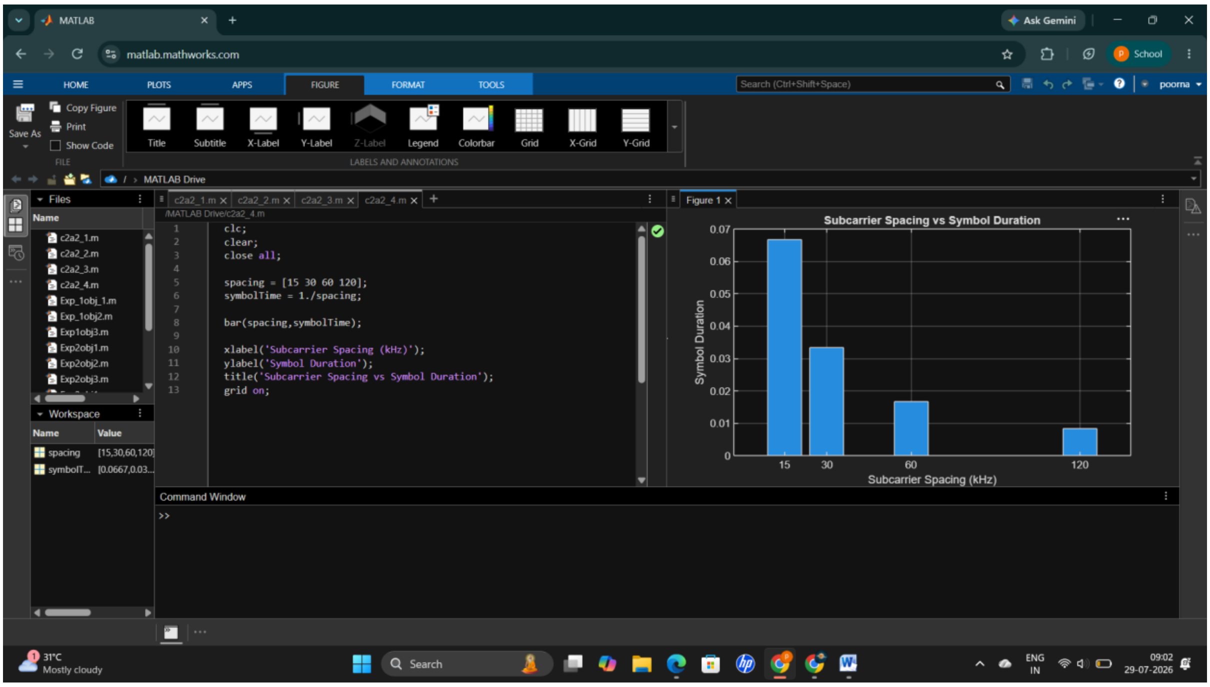
```
clc;
clear;
close all;

spacing = [15 30 60 120];
symbolTime = 1./spacing;

bar(spacing,symbolTime);

xlabel('Subcarrier Spacing (kHz)');
ylabel('Symbol Duration');
title('Subcarrier Spacing vs Symbol Duration');
grid on;
```

Output:-



Result:-

Higher subcarrier spacing results in shorter symbol duration, making it suitable for low-latency 5G communication.

5. Simulation of Propagation Loss under Different Frequencies.

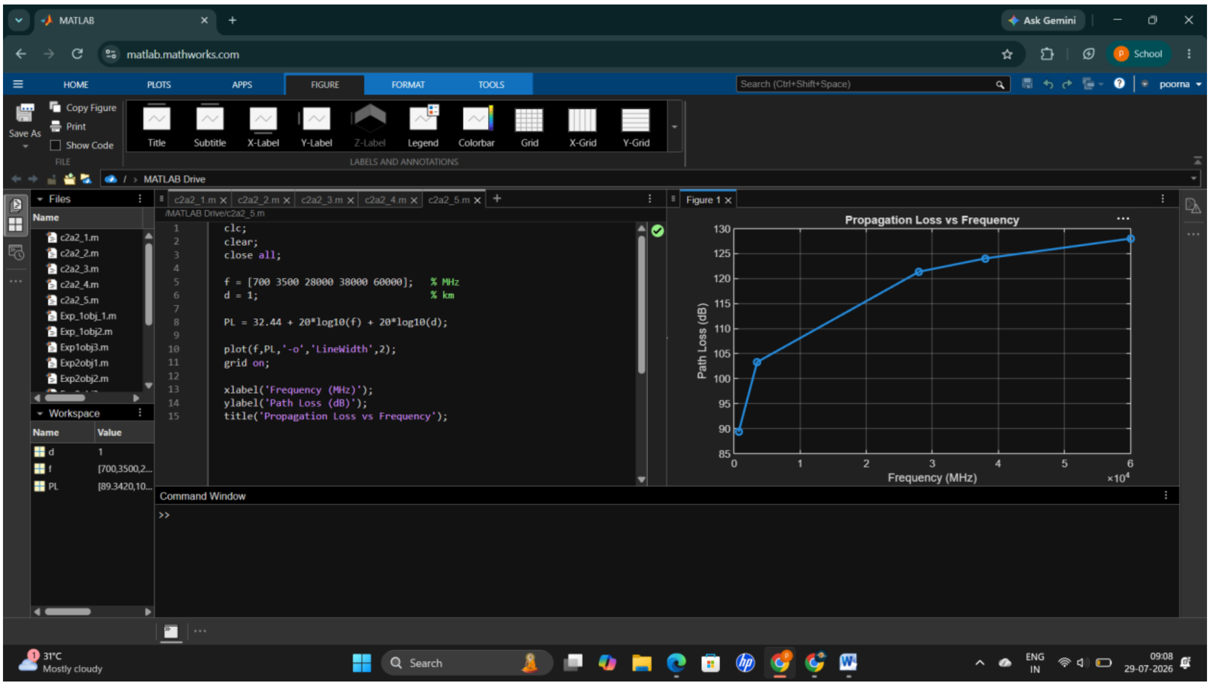
Aim:-

To simulate propagation loss at different operating frequencies.

MATLAB Code:-

```
clc;
clear;
close all;
f = [700 3500 28000 38000 60000];   % MHz
d = 1;                              % km
PL = 32.44 + 20*log10(f) + 20*log10(d);
plot(f,PL,'-o','LineWidth',2);
grid on;
xlabel('Frequency (MHz)');
ylabel('Path Loss (dB)');
title('Propagation Loss vs Frequency');
```

Output:-



Result:-

Propagation loss increases with operating frequency. Higher-frequency bands such as mmWave experience greater attenuation than lower-frequency bands, requiring techniques like beamforming and small cells for reliable communication.